How We Pay Matters?

Payment Modality, Gender, and Spending Behavior in Cash Transfer Programs

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Project Question: How does receiving cash transfers through mobile wallets, compared to physical cash, affect household spending patterns across essential versus discretionary purchases in Indonesia, and does this effect differ by gender?

Topic and Motivation

This research project addresses the intersection of digital financial inclusion and social protection policy, specifically examining how the delivery of cash transfer programs influences household economic behavior. As governments worldwide increasingly digitalize social assistance programs, understanding whether the method of transfer delivery (mobile wallets or physical cash) affects how recipients allocate resources has significant implications for program design and poverty alleviation outcomes.

The shift to digital transfers matters for several reasons. First, mobile wallet-based transfers may reduce transaction costs and improve transparency and record-keeping for recipients. Second, digital payments could alter spending patterns by creating psychological framing effects. Specifically, money held in a mobile wallet may be perceived or spent differently than physical cash.[1] Third, if delivery mechanisms differentially affect men and women's spending decisions, this could have implications for child welfare outcomes, given established evidence that women's control over resources tend to increase expenditure on children's needs.[2]

Indonesia provides an ideal setting for this research project. The country operates one of the largest social assistance programs in the developing world, with cash transfers reaching over 10 million households through programs like Program Keluarga Barapan and Bantuan Pangan Non Tunai.[3] Simultaneously, the Indonesian government has been progressively shifting toward digital payment mechanisms as part of its broader financial inclusion agenda.[4]

Currently, Indonesia's cash transfer programs use mixed delivery mechanisms: some beneficiaries receive physical cash at distribution points, others receive transfers through bank accounts, and an increasing number receive funds via mobile wallets.[5] While existing research has examined cash versus in-kind transfers and mobile money adoption separately, there is limited experimental evidence on how cash transfer delivery channels affect spending behavior, particularly in Southeast Asia. Moreover, few studies have rigorously examined whether these effects vary by recipient gender, despite theoretical reasons to expect differences based on gendered financial decision-making patterns. This evidence gap is critical: if digital delivery mechanisms systematically alter spending patterns between essential and discretionary goods, this would have important welfare implications beyond efficiency gains that should inform social protection program design.

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[1] Thaler, Richard H. 1999. "Mental Accounting Matters." *Journal of Behavioral Decision Making* 12 (3): 183–206. https://doi.org/10.1002/(SICI)1099-0771(199909)12:3%3C183::AID-BDM318%3E3.0.CO;2-F.
[2] Duflo, Esther. 2003. "Grandmothers and Granddaughters: Old-Age Pensions and Intrahousehold Allocation in South Africa." The World Bank Economic Review 17, no. 1 (June): 1–25. https://academic.oup.com/wber/article-abstract/17/1/1/1676291.
[3] World Bank. 2019. Aspiring Indonesia—Expanding the Middle Class. World Bank, Washington, DC. https://doi.org/10.1596/33237.
[4] Darjana, and Yanti Rusmawati. 2025. "Performance of Digital Economic Indicators During the COVID-19 Pandemic: Payment System Case of Indonesia." Bulletin of Monetary Economics and Banking 28, no. 4: 595–608. https://bulletin.bmeb-bi.org/cgi/viewcontent.cgi?article=2513&context=bmeb.
[5] Palaon, Hilman. 2020. "Digitizing G2P Payments: Lessons from Indonesia." CGAP (blog). July 21, 2020. https://www.cgap.org/blog/digitizing-g2p-payments-lessons-indonesia.

Research Question

The question guiding this research is as follows: How does receiving cash transfers through mobile wallets, compared to physical cash, affect the allocation of household spending between essential purchases (food, education, healthcare) and discretionary purchases (entertainment, non-essential consumer goods) among beneficiaries of social assistance programs in Indonesia, and does this effect differ by the gender of the primary recipient?

The key variables to assess the research question as follows:

- **Treatment Variable (X):** Delivery mechanism of cash transfers (mobile wallet vs physical cash).
- **Moderator Variable:** gender of the primary transfer recipient.
- **Outcome (Y):** Household expenditure patterns measured as:
    - Share of total spending allocated to essential categories (food, education, healthcare)
    - Share of total spending allocated to discretionary categories (entertainment, non-essential consumer goods, durables, gifts)
    - Total expenditure levels by category
- **Population:** Low-income households receiving government cash transfers in Indonesia, specifically beneficiaries of the social assistance programs.

Initial Literature Scan

Current literature emphasizes the reduced opportunity costs and friction associated with mobile wallets. An unconditional cash transfer program in Niger found that deposits in mobile wallets compared to enveloped-cash increased consumption of protein-rich foods and induced a greater diversity in diet. The authors hypothesize that forgoing the transport costs required to collect cash both lead to greater time allocation and improved the economic bargaining power for women.[6]

Similarly, evidence on women's empowerment in Kenya found women who received mobile money transfers (again compared to cash), were more likely to independently invest in income-generating activity. The study found individual earnings for women increased by 31% compared to the transfer of physical cash.[7] These findings are in line with the literature which suggests that targeting cash transfers to women not only promote gender equity, but also increase the expenditure on higher quality foods.[8]

Households leverage their informal risk-sharing networks through mobile money. Those who use mobile money are more resilient to economic shocks than those who do not. An RCT in Kenya found evidence that those that do not use mobile money experience a 7% drop in consumption after an economic shock.[9]

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[6] Aker, J. C., Boumnijel, R., McClelland, A., & Tierney, N. 2016. Payment mechanisms and antipoverty programs: Evidence from a mobile money cash transfer experiment in Niger. Economic Development and Cultural Change, 65(1), 1–37. https://doi.org/10.1086/687578

[7] Greco, G., Gulesci, S., Prabhakar, P., & Sulaiman, M. 2025. *(Digital) cash transfers, privacy and women's empowerment: Evidence from Uganda*. BRAC Institute of Governance and Development. https://bigd.bracu.ac.bd/publications/digital-cash-transfers-privacy-and-womens-empowerment-evidence-from-uganda/

[8] Armand, A., Attanasio, O., Carneiro, P., & Lechene, V. 2020. The Effect of Gender-Targeted Conditional Cash Transfers on Household Expenditures: Evidence from a Randomized Experiment. *Economic Journal*, *130*(631), 1875–1897. https://doi-org.proxy.library.georgetown.edu/10.1093/ej/ueaa056

[9] Alinaghi, N. & Reed, W. 2018. *Risk sharing and transaction costs: Evidence from Kenya's mobile money revolution, a replication study of Jack and Suri (2014) — Revised replication plan*. International Initiative for Impact Evaluation (3ie). https://www.3ieimpact.org/sites/default/files/2018-12/alinaghi_revised_replication_plan.pdf

Discussions about mobile money in developed countries have focused on the "cashless effect," an increased willingness-to-pay induced by using cashless forms of payment. The theory suggests physical forms of money are psychologically more salient, while digital forms are less salient, meaning that consumers become less concerned about amount when they use digital forms of payment.[10] Results have been mixed, with research finding positive, null, and negative results.

Theory of Change

The key assumption undergirding our hypothesis is that the modality of a cash transfer changes the properties of money, specifically how people access, store, observe, and spend their resources. These changes operate through three potential mechanisms.

First, regarding the observability of expenses, mobile money transactions are less physically salient than cash transactions. This generates a "cashless effect,[11]" whereby digital payments reduce the psychological friction associated with spending money at the moment of purchase. This effect would suggest that mobile money could result in an increase in discretionary or impulsive purchases, while the observability of cash could lead consumers to spend more on essentials. However, given that many mobile wallets notify users about their spending and balance, the "cashless effect" may be ambiguous as mobile wallets may nudge users to budget and spend frugally.

Secondly, time and transaction costs would be lower with digital transfers that reduce travel and waiting time associated with collecting and safeguarding cash. Lower transaction costs could increase effective transfer value and enable greater investment in essentials, particularly if reduced frictions improve the efficiency of household resource allocation. If digital wallets also lower the cost of saving and managing money, this could result in higher savings.[12]

Lastly, mobile money could critically affect intra-household control and bargaining power, influencing who effectively controls the funds they receive. Mobile wallets registered to women are more likely to secure privacy and autonomy. With greater control over resources for women, household spending may shift toward categories such as food quality, children's schooling, and health.[13]

Together, these channels generate two competing but not mutually exclusive pathways to the final outcome. The "cashless effect" pathway predicts higher discretionary spending due to reduced payment salience and increased transaction ease. The "control and budgeting pathway" predicts greater spending on essentials if digital delivery enhances financial autonomy and record-keeping. The gender of the recipient is expected to moderate these effects, as shifts in control may differentially alter intra-household allocation decisions.

Our key assumption is that transfer recipients must be able to access and use the mobile wallet reliably and appropriately. Causal chains could break if most recipients immediately cash out so the wallet becomes equivalent to cash, if agent networks and fees make wallet use impractical, or if spending is

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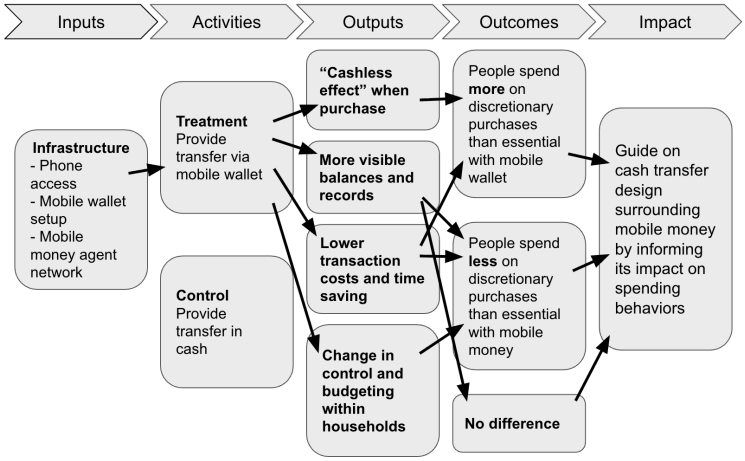
[10]Schomburgk, L. Belli, A. & Hoffmann, A. 2024. Less cash, more splash? A meta-analysis on the cashless effect. Journal of Retailing, Volume 100, Issue 3, 2024, Pages 382-403, ISSN 0022-4359, https://doi.org/10.1016/j.jretai.2024.05.003.

[11] Ibid.

[12] See Alinaghi & Reed, 2018

[13] Aker, Jenny C., Rachid Boumnijel, Amanda McClelland, and Niall Tierney. 2016. Payment Mechanisms and Anti-Poverty Programs: Evidence from a Mobile Money Cash Transfer Experiment in Niger. J-PAL. https://www.povertyactionlab.org/sites/default/files/research-paper/4277_Payments-Mechanism-and-anti-poverty-programs_Aker_Nov2016.pdf

measured differently across arms. If assignment is not fully randomized, differences in wealth, urbanization, digital literacy, or local market conditions between groups could confound estimated effects on spending composition.



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